Financial Derivatives
ES32079

Lecture 3

Interest rates and bonds

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Structure of this lecture

Here we cover the following topics:

1. Interest Rates
2. Discounting and Compounding
3. Bond Pricing
4. Bootstrapping the Zero Curve
5. Forward Rates
6. Forward Rate Agreements
7. Duration and Convexity
8. Summary

Reference: Hull chapter 4.

Why Do Interest Rates Matter?

Interest rates enter almost every derivatives-pricing problem because cash flows occurring at different dates cannot be compared directly.

- They determine how future cash flows are discounted.
- They affect bond prices, forward prices, swaps, options, and other derivatives.
- They influence borrowing, saving, investment, exchange rates, and monetary policy.
- Different rates reflect both the value of time and, in many cases, credit, liquidity, and collateral risk.

Central question

How much is a payment at a future date worth today?

Interest Rates Are Not All the Same

Government bond rates Yields on debt issued by governments; these vary by currency, maturity, liquidity, and sovereign credit risk.

Policy rates Rates set or targeted by central banks as part of monetary policy.

Repo rates Rates on secured borrowing in which securities serve as collateral.

Reference rates Transaction-based benchmarks used in loans and derivatives, such as SONIA for sterling and SOFR for US dollars.

Risk-free rate

The risk-free rate is a modelling ideal: the return on an investment with no default risk. In modern derivatives markets, overnight-indexed rates are often used to construct discount curves, subject to the contract and collateral arrangements.

Reference Rates After LIBOR

LIBOR is now a historical benchmark: all LIBOR settings have permanently ceased.

- **SONIA** is the Sterling Overnight Index Average. It reflects rates paid on eligible overnight sterling transactions.
- **SOFR** is the Secured Overnight Financing Rate. It measures the cost of overnight borrowing collateralised by US Treasury securities.
- Overnight rates may be compounded over a period to determine the interest payable on a loan or derivative.

Teaching convention

Later FRA examples use a generic market reference rate R_M. Older textbooks may label the same quantity "LIBOR."

Sources: Financial Conduct Authority; Bank of England; Federal Reserve Bank of New York.

What Does "10% Per Annum" Mean?

The quotation is incomplete unless its compounding convention is stated.

One-year investment

Invest \$100 at a nominal annual rate of 10%.

- Annual compounding: $100 \times 1.10 = \$110.00$
- Semiannual compounding: $100(1 + 0.10/2)^2 = \$110.25$
- Continuous compounding: $100e^{0.10} = \$110.52$

The quoted rate alone is therefore not enough to determine the terminal value.

Compounding m Times per Year

If A is invested for T years at a nominal annual rate R_m, compounded m times per year, its future value is

$$\boxed{FV = A\left(1 + \frac{R_m}{m}\right)^{mT}}.$$

The corresponding present value is

$$\boxed{PV = FV\left(1 + \frac{R_m}{m}\right)^{-mT}}.$$

Example

At 8% per annum with quarterly compounding, the two-year value of \$1,000 is

$$1{,}000\left(1 + \frac{0.08}{4}\right)^{8} = \$1{,}171.66.$$

The Effect of Compounding Frequency

Compounding	m	Value of \$100 after one year
Annual	1	\$110.00
Semiannual	2	\$110.25
Quarterly	4	\$110.38
Monthly	12	\$110.47
Daily	365	\$110.52

Effective annual rate

The effective annual rate generated by R_m is

$$R_{\text{eff}} = \left(1 + \frac{R_m}{m}\right)^m - 1.$$

Thus 10% with semiannual compounding has an effective annual rate of $1.05^2 - 1 = 10.25\%$.

Continuous Compounding

The exponential function is defined by

$$e^x = \lim_{m\to\infty}\left(1+\frac{x}{m}\right)^m.$$

As the compounding frequency becomes arbitrarily large,

$$\lim_{m\to\infty}\left(1+\frac{r}{m}\right)^{mT} = e^{rT}.$$

Therefore, under continuous compounding,

$$\boxed{FV = Ae^{rT}}, \qquad \boxed{PV = FV\, e^{-rT}}.$$

Sign check

Compounding moves money forward with e^{rT}; discounting moves it backward with e^{-rT}.

Converting Interest-Rate Conventions

Equivalent rates generate the same accumulation over the same period:

$$e^r = \left(1 + \frac{R_m}{m}\right)^m.$$

Hence,

$$\boxed{R_m = m\left(e^{r/m} - 1\right)}, \qquad \boxed{r = m \ln\left(1 + \frac{R_m}{m}\right)}.$$

Example

A nominal rate of 10% with semiannual compounding is equivalent to

$$r = 2\ln(1.05) = 0.09758,$$

or 9.758% with continuous compounding.

Compounding Check

An investment grows from \$5,000 to \$5,800 in 18 months.

1. What is the continuously compounded annual rate?
2. What nominal annual rate with semiannual compounding is equivalent?

Solution

$$r = \frac{1}{1.5} \ln\left(\frac{5{,}800}{5{,}000}\right) = 10.15\%.$$

The equivalent semiannual rate is

$$R_2 = 2\left(e^{0.1015/2} - 1\right) = 10.41\%.$$

Zero Rates and Discount Factors

The T-year zero rate is the return on an investment that makes no intermediate payments and pays only at time T.
With a continuously compounded zero rate $R(T)$, the discount factor is

$$\boxed{P(0,T) = e^{-R(T)T}}.$$

The present value of a cash flow C_T is

$$PV(C_T) = C_T P(0,T) = C_T e^{-R(T)T}.$$

Example

If the five-year zero rate is 5%, then

$$100e^{0.05\times 5} = \$128.40, \qquad 100e^{-0.05\times 5} = \$77.88.$$

A Bond Is a Package of Cash Flows

A coupon bond normally pays:

- coupons at specified dates; and
- face value, also called principal or par value, at maturity.

If cash flow C_i is paid at time t_i, its price is

$$B = \sum_{i=1}^{n} C_i P(0, t_i) = \sum_{i=1}^{n} C_i e^{-R(t_i)t_i}.$$

Key principle

Each cash flow must be discounted using the zero rate for its own maturity. Using one rate for every cash flow produces a yield measure, not a no-arbitrage price from the zero curve.

Pricing a Coupon Bond from Zero Rates

Consider a two-year bond with face value \$100 and a 6% annual coupon paid semiannually. Each coupon is \$3.

Time t_i	0.5	1.0	1.5	2.0
Cash flow C_i	3	3	3	103
Zero rate	5.0%	5.8%	6.4%	6.8%

Assuming continuous compounding,

$$\begin{aligned} B &= 3e^{-0.05\times 0.5} + 3e^{-0.058\times 1} \\ &\quad + 3e^{-0.064\times 1.5} + 103e^{-0.068\times 2} \\ &= \$98.39. \end{aligned}$$

Bond Yield

The yield to maturity y is the single discount rate that reproduces the observed bond price:

$$B = \sum_{i=1}^{n} C_i e^{-yt_i}.$$

For the preceding bond,

$$98.39 = 3e^{-0.5y} + 3e^{-y} + 3e^{-1.5y} + 103e^{-2y}.$$

Solving numerically gives

$$\boxed{y \approx 6.76\%}.$$

Do not confuse

The zero curve contains one rate for each maturity. Yield to maturity compresses all of the bond's cash flows and the observed price into one internal rate of return.

Price, Coupon Rate, and Yield

For a standard fixed-rate bond:

Relationship	Bond price
Coupon rate $>$ yield	Above par
Coupon rate $=$ yield	At par
Coupon rate $<$ yield	Below par

Concept check

The example bond has a 6% coupon rate and a yield of approximately 6.76%. Should its price be above or below \$100?

It should trade below par, consistent with its price of \$98.39.

Par Yield

The par yield for maturity T is the coupon rate that makes a bond's price equal to its face value.
Let m be the number of coupon payments per year, let

$$A = \sum_{i=1}^{n} P(0, t_i)$$

be the present value of receiving \$1 on every coupon date, and let $d = P(0, T)$. For face value \$100,

$$A\frac{c}{m} + 100d = 100.$$

Therefore,

$$\boxed{c = \frac{100m(1-d)}{A}}.$$

Here c is the annual coupon in dollars on \$100 face value, numerically equal to the annual coupon rate in percent.

Par Yield: Numerical Example

For the two-year zero curve,

$$d = e^{-0.068\times 2} = 0.87284$$

and

$$A = e^{-0.05\times 0.5} + e^{-0.058\times 1} + e^{-0.064\times 1.5} + e^{-0.068\times 2} = 3.70027.$$

With two coupons per year,

$$c = \frac{100 \times 2 \times (1 - 0.87284)}{3.70027} = 6.87.$$

Conclusion

The two-year par yield is approximately 6.87% per annum with semiannual coupon payments.

What Is Bootstrapping?

Zero rates can be inferred from prices of traded instruments.

- A zero-coupon instrument reveals its maturity's discount factor directly.
- A coupon bond contains several cash flows.
- Once earlier discount factors are known, subtract the present value of the earlier coupons and solve for the next discount factor.
- Repeat sequentially along the maturity structure.

Why the name?

Each newly determined zero rate is used to obtain the next one.

Bootstrap Data

All bonds have face value \$100. Coupon-bearing bonds pay half the stated annual coupon every six months.

Maturity (years)	Annual coupon (\$)	Price (\$)
0.25	0	97.5
0.50	0	94.9
1.00	0	90.0
1.50	8	96.0
2.00	12	101.6

Starting point

The first three instruments are zero-coupon bonds, so their zero rates can be calculated directly.

Bootstrap Step 1: Zero-Coupon Instruments

For a zero-coupon bond with price B, face value 100, and maturity T,

$$B = 100e^{-R(T)T} \quad \Longrightarrow \quad R(T) = \frac{1}{T}\ln\left(\frac{100}{B}\right).$$

Therefore,

$$R(0.25) = \frac{1}{0.25}\ln\left(\frac{100}{97.5}\right) = 10.127\%,$$

$$R(0.50) = \frac{1}{0.50}\ln\left(\frac{100}{94.9}\right) = 10.469\%,$$

$$R(1.00) = \ln\left(\frac{100}{90.0}\right) = 10.536\%.$$

Bootstrap Step 2: The 1.5-Year Rate

The 1.5-year bond pays \$4 after 0.5 years, \$4 after one year, and \$104 after 1.5 years. Let $R(1.5)$ be the unknown zero rate:

$$4e^{-0.10469\times 0.5} + 4e^{-0.10536\times 1} + 104e^{-1.5\cdot R(1.5)} = 96.$$

After subtracting the present value of the first two coupons,

$$e^{-1.5\cdot R(1.5)} = 0.85196.$$

Thus,

$$\boxed{R(1.5) = -\frac{\ln(0.85196)}{1.5} = 10.681\%}.$$

Bootstrap Step 3: The 2-Year Rate

The two-year bond pays semiannual coupons of \$6. Let $R(2)$ denote the unknown two-year zero rate:

$$101.6 = 6e^{-0.10469\times 0.5} + 6e^{-0.10536\times 1} + 6e^{-0.10681\times 1.5} + 106e^{-2\times R(2)}.$$

Solving gives

$$\boxed{R(2) = 10.808\%}.$$

Check the method

Why could the two-year rate not be calculated before the 1.5-year rate?

The Bootstrapped Zero Curve

Maturity (years)	Continuously compounded zero rate
0.25	10.127%
0.50	10.469%
1.00	10.536%
1.50	10.681%
2.00	10.808%

Between observed maturities

An interpolation method must be chosen. Linear interpolation of zero rates is one simple classroom convention, but market curve construction may instead interpolate discount factors, log discount factors, or forward rates and may use more sophisticated smoothing methods.

What Is a Forward Interest Rate?

A forward rate is the future interest rate implied today by current zero rates.

If R_1 and R_2 are continuously compounded zero rates for maturities $T_1 < T_2$, no arbitrage requires

$$e^{R_2 T_2} = e^{R_1 T_1} e^{R_F (T_2 - T_1)}.$$

Hence,

$$\boxed{R_F(T_1, T_2) = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}}.$$

Equivalently, using discount factors,

$$e^{-R_F (T_2 - T_1)} = \frac{P(0, T_2)}{P(0, T_1)}.$$

Forward-Rate Example

Suppose the continuously compounded zero rates are 3% for one year and 4% for two years. The one-year rate beginning one year from today is

$$R_F(1,2) = \frac{0.04 \times 2 - 0.03 \times 1}{2 - 1} = 5\%.$$

Both investment strategies therefore produce the same terminal value:

$$100e^{0.03\times 1}e^{0.05\times 1} = 100e^{0.04\times 2} = \$108.33.$$

Interpretation

The 5% forward rate is implied by today's curve. It is not a guarantee that the one-year market rate observed in one year's time will equal 5%.

Forward Rates from a Zero Curve

Year n	n-year zero rate	Forward rate for year n
1	3.0%	—
2	4.0%	5.0%
3	4.6%	5.8%
4	5.0%	6.2%
5	5.3%	6.5%

For example, the year-four forward rate is

$$R_F(3,4) = \frac{0.05 \times 4 - 0.046 \times 3}{4-3} = 6.2\%.$$

Your turn

Use the zero rates to verify the 5.8% forward rate for year three.

What Is a Forward Rate Agreement?

A forward rate agreement (FRA) is an over-the-counter contract that fixes the interest rate applied to a notional principal during a future period $[T_1, T_2]$.

- The principal is normally not exchanged.
- The parties exchange only an interest-rate difference.
- One party receives fixed and pays the future market reference rate.
- The other party pays fixed and receives the market rate.
- Settlement normally occurs at T_1, so the T_2 difference is discounted back to T_1.

Economic purpose

A borrower can use an FRA to protect against rising rates; a lender can use one to protect against falling rates.

FRA Notation and Payoff

Define

- L: notional principal;
- R_K: fixed rate agreed in the FRA;
- R_M: market reference rate observed at T_1;
- $\delta = T_2 - T_1$: year fraction for the underlying period.

For the party that **receives fixed and pays floating**, the cash flow at T_2 is

$$L(R_K - R_M)\delta.$$

If settlement occurs at T_1, its value is

$$\boxed{\frac{L(R_K - R_M)\delta}{1 + R_M\delta}}.$$

The opposite party receives the negative of this amount.

FRA Example: Protecting a Lending Rate

A company enters an FRA to receive a fixed rate of 4% on \$100 million for a three-month period beginning in three years. Thus $\delta = 0.25$.
If the market reference rate at T_1 is 4.5%, the receive-fixed party's cash flow at T_2 is

$$100{,}000{,}000 \times (0.04 - 0.045) \times 0.25 = -\$125{,}000.$$

Settled at T_1, the amount is

$$\frac{-125{,}000}{1 + 0.045 \times 0.25} = -\$123{,}609.$$

Interpretation

The FRA loss offsets the benefit of being able to lend at the higher market rate. Together, the loan and FRA lock in approximately 4%.

Which Side of the FRA?

For each exposure, decide whether the firm should pay fixed or receive fixed.

1. A company will borrow £10 million in six months and fears that rates will rise.
2. An investor will lend £10 million in six months and fears that rates will fall.

Answer

1. The future borrower should **pay fixed and receive floating**. If rates rise, the FRA gain offsets the higher borrowing cost.
2. The future lender should **receive fixed and pay floating**. If rates fall, the FRA gain offsets the lower lending income.

Bond Price as a Function of Yield

Using a continuously compounded yield y,

$$\boxed{B(y) = \sum_{i=1}^{n} C_i e^{-yt_i}}.$$

Differentiating,

$$\frac{dB}{dy} = -\sum_{i=1}^{n} t_i C_i e^{-yt_i} < 0.$$

Therefore, bond price and yield move in opposite directions.

Prediction

Which should be more sensitive to a rate change: a two-year zero-coupon bond or a ten-year zero-coupon bond? Why?

Macaulay Duration under Continuous Compounding

Duration is a present-value-weighted average payment time:

$$\boxed{D = \frac{1}{B}\sum_{i=1}^{n} t_i C_i e^{-yt_i}}.$$

The weight of cash flow i is

$$w_i = \frac{C_i e^{-yt_i}}{B}, \qquad \sum_{i=1}^{n} w_i = 1,$$

so $D = \sum_i t_i w_i$.

- A zero-coupon bond's duration equals its maturity.
- A coupon bond's duration is shorter than its maturity because some cash flows arrive earlier.

Duration and Price Sensitivity

Because

$$\frac{dB}{dy} = -BD,$$

a small yield change Δy produces

$$\boxed{\frac{\Delta B}{B} \approx -D\Delta y}.$$

Units matter

One basis point is

$$1 \text{ bp} = 0.01\% = 0.0001.$$

A 10-basis-point increase means $\Delta y = +0.001$, not 0.1.

Longer duration means greater percentage price sensitivity to the same small change in yield.

Duration Example

Consider a three-year, 10% coupon bond paying semiannually. Its continuously compounded yield is 12%.

Time	Cash flow	Present value	$t_i \times$weight
0.5	5	4.709	0.025
1.0	5	4.435	0.047
1.5	5	4.176	0.066
2.0	5	3.933	0.083
2.5	5	3.704	0.098
3.0	105	73.256	2.333
Total	130	94.213	2.653

Thus $B = 94.213$ and $D = 2.653$ years.

Using Duration to Estimate a Price Change

Suppose the yield rises by 10 basis points:

$$\Delta y = +0.001.$$

Duration predicts

$$\frac{\Delta B}{B} \approx -2.653 \times 0.001 = -0.002653,$$

so

$$\Delta B \approx -94.213 \times 2.653 \times 0.001 = -0.250.$$

The estimated new price is

$$94.213 - 0.250 = 93.963.$$

Direct repricing at $y = 12.1\%$ gives approximately 93.963, so duration is very accurate for this small change.

Why Duration Is Only an Approximation

Duration replaces the curved price-yield relationship with a tangent line.

- It is accurate for small yield changes.
- Its error grows as the yield change becomes larger.
- For an ordinary option-free bond, the true price-yield relationship is convex.

Yield increase	Exact price	Duration estimate	Error
10 bp	93.963	93.963	approximately 0.000
100 bp	91.749	91.714	0.035
200 bp	89.354	89.214	0.140

Convexity

Convexity measures the curvature of the price-yield relationship:

$$\boxed{C = \frac{1}{B}\frac{d^2B}{dy^2} = \frac{1}{B}\sum_{i=1}^{n} t_i^2 C_i e^{-yt_i}}.$$

The second-order approximation is

$$\boxed{\frac{\Delta B}{B} \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2}.$$

For the example bond,

$$C = 7.570.$$

Duration Plus Convexity

For a 100-basis-point yield increase, $\Delta y = 0.01$. The duration-convexity estimate is

$$\frac{\Delta B}{B} \approx -2.653 \times 0.01 + \frac{1}{2} \times 7.570 \times (0.01)^2.$$

Therefore,

$$B_{\text{estimated}} = 94.213\,[1 - 0.02653 + 0.0003785] = 91.749.$$

The exact repriced value is also approximately 91.749, compared with the duration-only estimate of 91.714.

Interpretation

Convexity corrects much of the error created by the linear duration approximation.

Duration, Convexity, and Immunisation

Matching the duration of assets and liabilities can reduce sensitivity to small parallel yield-curve shifts.

- Convexity improves protection against larger parallel shifts.
- Duration and convexity change as time passes and yields move.
- Matching them does not eliminate exposure to non-parallel curve movements.
- Credit spreads, liquidity, optionality, and model risk may remain.

Risk-management lesson

Duration and convexity describe particular dimensions of interest-rate risk; they do not make a portfolio riskless.

Lecture Summary

- Interest-rate quotations require a compounding convention.
- Zero rates discount individual cash flows at their respective maturities.
- Bond yield is a single internal rate; par yield is the coupon rate that makes a bond trade at par.
- Bootstrapping extracts discount factors sequentially from market prices.
- Forward rates are implied by today's zero curve.
- FRAs fix a rate for a future borrowing or lending period.
- Duration measures first-order price sensitivity; convexity provides a second-order correction.

Unifying idea

Represent every instrument as dated cash flows, discount those cash flows consistently, and then examine how the value changes when rates move.